To find the position of m_3 the Pythagoras theorem is applied. As the ΔDBC is a right angle triangle,

$$BC^2 = CD^2 + DB^2$$

$$CD^2 = BC^2 - DB^2$$

$$CD^2 = 1^2 - \left(\frac{1}{2}\right)^2 = 1 - \left(\frac{1}{4}\right) = \frac{3}{4}$$

$$CD = \frac{\sqrt{3}}{2}$$

The position of mass m_3 is

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \text{ or } \left(0.5, 0.5\sqrt{3}\right)$$

X Coordinate of centre of mass,

$$x_{CM} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

$$x_{CM} = \frac{(1\times 0) + (2\times 1) + (3\times 0.5)}{1+2+3} = \frac{3.5}{6}$$

$$x_{CM} = \frac{7}{12}\text{ m}$$

Y Coordinate of centre of mass,

$$y_{CM} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$$

$$y_{CM} = \frac{(1\times 0) + (2\times 0) + (3\times 0.5\times \sqrt{3})}{1+2+3} = \frac{1.5\sqrt{3}}{6}$$

$$y_{CM} = \frac{\sqrt{3}}{4}\text{ m.}$$

∴ The coordinates of centre of mass G $\left(x_{CM}, y_{CM}\right)$ is $\left(\frac{7}{12}, \frac{\sqrt{3}}{4}\right)$

2. An electron of mass $9\times10^{-31}\,kg$ revolves around a nucleus in a circular orbit of radius 0.53 Å. What is the angular momentum of the electron? (Velocity of electron is, $v = 2.2\times10^{6}\,m\,s^{-1}$)

Solution:

Mass of the electron, $m = 9\times10^{-31}\,kg$

Radius of the electron, r = 0.53 Å = $0.53\times10^{-10}\,m$

Velocity of the electron, v = $2.2\times10^{6}\,m\,s^{-1}$

Angular momentum of electron is, L = I ω

Electron is considered as a point mass. Hence, its moment of inertia is, $I = m\,r^2$

The relation, $\omega = \frac{v}{r}$ could be used.

Angular momentum, $L = mr^2\times\frac{v}{r}$

= mvr

$$= 9.1\times10^{-31}\times2.2\times10^{6}\times0.53\times10^{-10}$$

$$L = 1.06\times10^{-34}\,kg\,m^2\,s^{-1}$$

3. A solid sphere of mass 20 kg and radius 0.25 m rotates about an axis passing through the centre. What is the angular momentum if the angular velocity is 5 rad s^{-1}

Solution:

Mass of the sphere, m = 20 kg

Radius r = 0.25 m

Appendix-1.indd 284 28-01-2020 14:14:06